

Evaluating Resource Adequacy Using Hourly 40 Weather-Year Data Under High VRE
Penetration for the Luzon and Visayas Grids in Alignment with the Power Development Plan
(2023-2050)

Undergraduate Student Project

by

Rafa Judah D. Tirones
2022-07684
BS Electrical Engineering

Ryan Jay C. Vico
2020-11990
BS Electrical Engineering

Adviser:

Adonis Emmanuel DC. Tio

University of the Philippines, Diliman

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Abstract

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The Philippines is pursuing a major energy transition, targeting a renewable energy share of 35% by 2030 and 50% by 2040, as detailed in the Power Development Plan (PDP) 2023–2050. This rapid integration of variable renewable energy (VRE) sources presents significant challenges to long-term resource adequacy. The PDP’s current planning methodology for the Philippines does not account for how unpredictable weather patterns can behave over long periods; specifically, how wind and solar output can drop significantly for extended times. This omission may lead to a systematic underestimation of the capacity needed to ensure supply adequacy during rare but high-impact challenging weather years. This research adapts the net load profiling from Ullmark et al. to the Philippine grid. Using 40 years (1985–2024) of hourly meteorological data from the MERRA-2 reanalysis dataset, this study generates multi-decade net-load fingerprints (recurrence matrices) for the DOE’s 2025, 2030, 2035, 2040, 2045, and 2050 VRE expansion scenarios. By identifying the challenging weather years from history, the study determines the necessary mix of dispatchable generation and long-term storage. This study provides a multi-decade and variability-driven adequacy assessment of the PDP 2023–2050. Ultimately, this study specifically assesses whether the planned installed capacity is sufficient to meet demand, accounting for the inherent variability of solar and wind resources, even during challenging years characterized by extreme weather events. The findings will identify and quantify any adequacy gap between the PDP’s planned capacity and the system’s reliability requirements, offering critical data-driven recommendations for national energy planners.

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Chapter 1

Introduction

Resource adequacy is a fundamental concept in power system planning that refers to the ability of supply-side and demand-side resources to meet the aggregate electrical demand, including system losses, at all times [1]. This assessment is crucial for ensuring that an electricity system maintains reliable service to consumers while balancing the economic considerations of capacity investments [2]. Resource adequacy is typically quantified using probabilistic metrics such as Loss of Load Expectation (LOLE), which measures the expected number of time periods during which system resources are insufficient to meet demand, and Expected Unserved Energy (EUE), which calculates the total amount of energy that cannot be served due to capacity shortfalls [3]. The widely-accepted reliability threshold of “1 day in 10 years” for LOLE has served as a benchmark for power system planners globally, indicating that during a decade of operations, a system is expected to experience shortfall events on only a single day [2].

The Power Development Plan (PDP) 2023-2050 has established ambitious renewable energy targets, aiming for a 35% share in the power generation mix by 2030 and 50% by 2040 [4]. These targets represent a transformative shift from the country’s current energy landscape, where coal-fired power plants dominate at approximately 59.6% of gross generation, followed by renewable energy at 22.1% and natural gas at 16.0%. The country’s total non-coincidental peak demand reached 16,596 MW in 2022, representing a recovery from pandemic-induced demand reductions and a return to pre-pandemic growth trajectories. Projections indicate that peak demand will significantly increase to 68,483 MW by 2050, equivalent to an annual average growth rate of 5.2%, necessitating approximately a five-fold increase in installed capacity.

The integration of variable renewable energy (VRE) sources, particularly solar and wind power, introduces unprecedented challenges to maintaining resource adequacy [5]. Unlike conventional thermal generation,

VRE output is inherently dependent on weather conditions, creating variability across multiple timescales—from minute-to-minute fluctuations to seasonal and interannual variations [6]. The net load, defined as electricity demand minus variable renewable generation, must be continuously balanced by dispatchable resources such as thermal power plants, hydroelectric facilities, and energy storage systems [7]. As VRE penetration increases, the characteristics of net load variability fundamentally change, potentially creating extended periods where dispatchable capacity is insufficient to meet demand [8].

Despite the critical importance of resource adequacy assessment, there exists a significant gap in research specifically addressing the Philippine context. Existing energy system models for the Philippines, such as the LEAP-based models developed for the Luzon grid, have primarily focused on generation expansion planning, low-carbon policy assessment, and energy storage integration without comprehensively addressing multi-decadal weather variability [9]. The recently completed Mindanao-Visayas Interconnection Project (MVIP), which established the first unified national grid connecting all three major island grids, creates new opportunities and challenges for system-wide resource adequacy assessment [4]. However, current planning methodologies employed in the PDP do not systematically account for how unpredictable weather patterns can behave over extended periods, particularly how wind and solar output can drop significantly for prolonged durations during challenging weather-years.

International studies have demonstrated that focusing on single weather years in capacity expansion modeling can lead to systematic misrepresentation of investment requirements for generation and storage capacity [10]. Research by Ullmark et al. (2025) on the North European electricity system has shown that the variability of wind and solar power outputs differs significantly between years, requiring high temporal resolution data covering multiple decades for satisfactory representation of variability [7]. Their proposed weather-year selection algorithm identifies representative sets of weather years that capture both typical conditions and particularly challenging high net-load events, enabling more robust resource adequacy assessment [11].

This research addresses this gap by adapting the net load fingerprinting by Ullmark et al. (2025) to evaluate resource adequacy for the Philippine grid under the VRE expansion scenarios outlined in the Philippine Energy Plan 2023-2050. By utilizing 40 years (1985-2024) of hourly meteorological data from the MERRA-2 reanalysis dataset, this study generates multi-decade net-load fingerprints for the Department of Energy's 2025, 2030, 2035, 2040, 2045, and 2050 VRE expansion scenarios. This enables determination of the necessary mix of dispatchable generation and long-term storage required to maintain grid reliability during rare but high-impact weather events. The findings will quantify any adequacy gap between the PDP's planned capacity and system reliability requirements, providing data-driven recommendations for national energy

planners navigating the transition to a high-renewable energy future.

Chapter 2

Related Work

This chapter details the different works related to resource adequacy, establishing the theoretical framework and policy context necessary for this study. This chapter discusses the current planning landscape through the Power Development Plan (PDP) 2023-2050. This also discusses the definition of Resource Adequacy in modern power systems. Lastly, this examines different existing works and solutions to identify the gaps that this study aims to address.

2.1 Power Development Plan 2023-2050

The Philippine Power Development Plan (PDP) 2023-2050 establishes the baseline planning methodology for the nation's energy transition toward renewable energy targets of 35% by 2030 and 50% by 2040 [4]. Its modeling follows a four-step approach: first, data gathering from historical and forecasted sources; second, development of three scenarios with associated constraints; third, execution of the Capacity Expansion Model (CEM) using PLEXOS with hourly time intervals from 2022 to 2050; and fourth, iterative refinement of model parameters and constraints. This methodology uses per-plant modeling of existing and committed power projects which optimizes new capacity additions across technologies: solar, wind, hydroelectric, geothermal, biomass, natural gas, battery storage, and nuclear, to meet projected demand each year.

In connection with the second step, the PDP examines three scenarios with increasing renewable ambition. The Reference Scenario aligns with the 35%-50% targets, projecting Luzon capacity expansion from 19,746 MW (2022) to 80,781 MW (2050), requiring 43,540 MW of new renewables, while Visayas capacity grows from 3,971 MW to 47,450 MW, needing 30,261 MW of new renewables [4]. Clean Energy Scenarios 1

and 2 (CES 1 and CES 2) pursue higher renewable penetration, with CES 1 assuming 19 GW of offshore wind and CES 2 assuming 50 GW, along with nuclear capacity ramping to 3,900 MW and 4,800 MW respectively. These scenarios require elevated battery storage, 11,846 to 14,846 MW on Luzon and 8,500 to 10,500 MW on Visayas, to manage increased intermittency. To present these planned capacity additions across all scenarios, figures 2.1a, 2.1b, 2.1c summarize them visually for Luzon, while figures 2.1d, 2.1e, 2.1f depict them for Visayas.

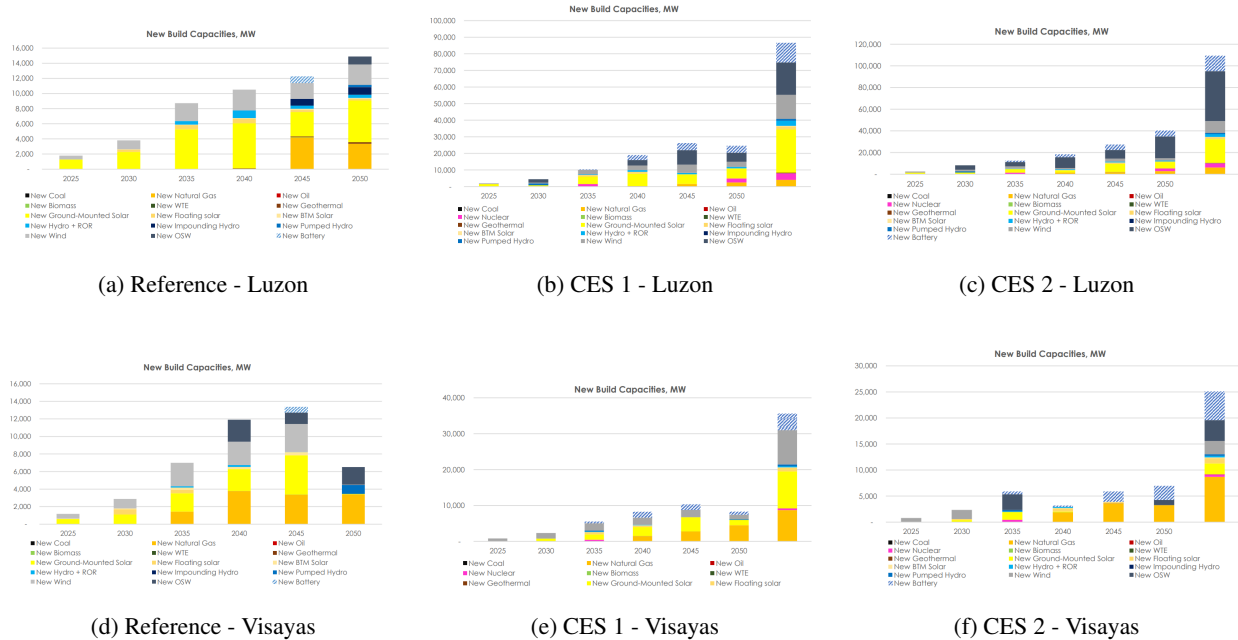


Figure 2.1: New Build Capacities per Technology per Scenario per Grid [4]

However, PDP does not systematically evaluate how multi-year periods of extreme weather affect resource adequacy and backup capacity requirements. While the model incorporates hourly weather profiles, it operates within a least-cost framework over the full planning horizon without explicitly testing system performance during historically challenging weather years. This approach may underestimate the dispatchable generation and storage needed when wind and solar output drops significantly for extended periods, particularly for the Luzon and Visayas grids.

2.2 Resource Adequacy

Resource adequacy assessment has evolved significantly as power systems transition toward higher penetrations of variable renewable energy. Traditional approaches relied on deterministic metrics such as Planning Reserve Margins (PRM), which calculates the percentage by which installed capacity exceeds peak de-

mand [3]. These methods assumed that capacity contributions could be represented by simple derating factors accounting for forced outage rates of conventional generators [12]. However, the increasing role of weather-dependent resources has necessitated more sophisticated probabilistic approaches.

Modern resource adequacy assessments employ Monte Carlo simulation methods to evaluate system reliability across thousands of potential scenarios representing different combinations of demand, renewable generation, and conventional unit availability [13]. Loss of Load Expectation (LOLE), measured in hours per year or days per year, has become the primary metric for quantifying adequacy risk [3]. The Expected Unserved Energy (EUE) metric provides complementary information about the magnitude of potential shortfalls, while Loss of Load Probability (LOLP) indicates the likelihood of shortfall events occurring.

The interconnection of multiple regions can enhance resource adequacy by enabling power sharing during stress events, but the benefits depend critically on the correlation of weather patterns and demand across interconnected zones [14]. For island systems like the Philippines, where interconnection capacity is limited by submarine cable constraints, the value of interconnections for adequacy enhancement may be bounded by transfer capacity limitations.

2.3 Existing Studies

Research on resource adequacy in the Philippine context remains limited, with existing studies primarily focusing on generation expansion planning without comprehensive treatment of interannual weather variability. In contrast, international research has substantially advanced understanding of resource adequacy assessment methodologies for modern power systems with variable renewable energy integration. Resource adequacy assessment is vital as it represents a tool for ensuring power system reliability as renewable penetration increases, with standardized metrics and probabilistic approaches forming the foundation of adequacy planning [15][16].

The Energy Systems Integration Group (ESIG) has understanding of resource adequacy criteria appropriate for electricity systems undergoing energy transitions [17]. ESIG research identifies fundamental limitations in traditional resource adequacy approaches based solely on LOLE metrics. Traditional capacity-based approaches, while effective for systems dominated by dispatchable thermal generation with high availability factors, prove inadequate for systems with substantial variable renewable penetration. Moreover, ESIG research demonstrates that modern adequacy assessment requires multi-metric approaches incorporating both LOLE and EUE metrics, with particular attention to tail risks and high-impact and rare events. The research highlights that damages from capacity shortfalls increase nonlinearly with event duration and

severity, necessitating differentiation between brief capacity shortfalls and extended energy deficiencies.

In addition, the Northwest Power and Conservation Council has developed probabilistic methods for adequacy assessment that explicitly incorporate the uncertainties most relevant to modern power systems [18]. Their LOLP methodology addresses the fundamental challenge that single-metric approaches fail to capture the full range of uncertainties affecting system adequacy. The framework demonstrates how incorporating runoff volume variability, temperature dependencies, generator forced outage rates, and wind generation variability into adequacy metrics produces more robust reliability assessments than deterministic reserve margin approaches. Furthermore, this shows that a five percent LOLP threshold, equivalent to the "once in 20 years" reliability standard, can be achieved through appropriate resource portfolio mix determination using probabilistic simulation.

Lastly, the weather-year selection methodology developed by Ullmark et al. (2025) represents a significant methodological advancement in representing interannual variability for capacity expansion modeling and adequacy assessment [7]. Their algorithm identifies sets of representative weather years that capture the full range of net-load variability across multi-decadal periods by analyzing both the amplitude and duration of net-load events. Their methodology enables cost-effective adequacy assessment across extended historical periods without requiring simultaneous modeling of all years. Its application to the North European electricity system demonstrates that well-selected representative weather years can accurately represent multi-decadal variability while substantially reducing computational requirements. Crucially, the research shows that accounting for extreme net-load events primarily increases requirements for peak thermal capacity and long-term energy storage, with different resources providing complementary adequacy benefits depending on event characteristics.

While these studies advance adequacy assessment, they either lack high-resolution meteorological integration or lack existing studies in the Philippines. This project adapts Ullmark's net-load fingerprinting to the Philippines, utilizing 40 years of hourly MERRA-2 data to assess the Department of Energy's PDP 2023–2050 scenarios.

Table 2.1: Summary of Related Works on Resource Adequacy

Reference	Focus	Tools	Method	Results
A Probabilistic Method to Assess Power Supply Adequacy for the Pacific Northwest [18]	Pacific Northwest adequacy standard	Genesys	Probabilistic LOLP with Monte Carlo; critical water conditions	5% LOLP threshold established.
New Resource Adequacy Criteria for the Energy Transition: Modernizing Reliability Requirements [17]	Modernizing adequacy criteria for energy transition	Conceptual framework	Multi-metric (LOLE, EUE); stress-testing extreme scenarios; cost-reliability trade-offs	LOLE insufficient alone; EUE preferred for energy-constrained systems; tail risks require deterministic testing.
Representing net load variability in electricity system capacity expansion models accounting for challenging weather-years [7]	Weather-year selection for capacity expansion	ECE model	Net-load recurrence matrix fingerprinting; iterative optimization; hand-picked extremes + optimized typical years	4–6 representative years capture 39-year net-load variability; extreme events drive thermal and storage investment.

2.4 Synthesis

The reviewed works reveal overlap between ambitious national policy targets and grid reliability. While the Power Development Plan (PDP) 2023–2025 establishes clear targets for high renewable energy penetration, this does not specifically address the interannual variability of renewable energy sources. The current planning methodology creates a gap of not being able to capture the risks of inherent weather-dependent power system.

International studies on resource adequacy highlight that as Variable Renewable Energy (VRE) penetration increases, traditional metrics lose their predictive value. This requires necessary shift toward assessments that can quantify the depth and duration of deficit events.

Thus, this study addresses these gaps by considering multi-decade weather-year data. This study also fo-

cuses on the Competitive Renewable Energy Zones to further assess the effects of the inherent intermittency of renewable energies such as solar and wind. This study, therefore, provides evaluation of long-term energy security by testing the net load against the installed capacity targets.

Chapter 3

Problem Statement and Objectives

3.1 Problem Statement

While the Philippine Energy Plan (PEP) 2020-2040 and the Power Development Plan (PDP) 2023-2050 set ambitious targets to transition the grid to a 50% renewable energy share by 2040, the methodology does not consider weather data. Furthermore, traditional probabilistic solutions also lack the temporal depth required to assess long-term energy security under high VRE penetration as they usually do not account for changing and extreme weather years. Consequently, reliance on these conventional methods risks underestimating the magnitude of potential supply deficits, leaving the grid vulnerable to prolonged outages during periods of weather-induced stress.

3.2 Objectives of the Project

The general objective of this study is to assess the sufficiency of the generation capacities of the Philippine power system under the capacity targets of the Power Development Plan (PDP) 2023-2050 by characterizing the interannual variability of renewable energy resources.

Specifically, this study aims to:

1. **Generate multi-decadal power generation profiles:** Develop representative solar and wind generation profiles spanning 40 years (1985-2024) using historical reanalysis weather data and spatial and hourly normalization based on Competitive Renewable Energy Zones (CREZ) for the Luzon and Visayas Grids.

2. **Characterize net load variability:** Analyze and characterize the full range of net load variability using techniques, including net load duration curves and recurrence matrices, to identify the frequency, severity, and characteristics of net load events.
3. **Evaluate generation sufficiency:** To determine the adequacy of the PDP's projected dependable capacities.
4. **Visualize reliability risks:** Present specific risk areas on the recurrence matrix heatmap to identify gaps in peaking capacity and energy storage.

3.3 Scopes and Limitations

The scope of this study is to focus on the resource adequacy of the Philippine power grid based on the Reference Scenario, Clean Energy Scenario 1 (CES 1), and Clean Energy Scenario 2 (CES 2) of the Department of Energy's Power Development Plan (2023–2050). The analysis covers the planning from 2025 to 2050 at five-year intervals.

To limit the scope of this study, the following limitations apply to the methodology and findings:

1. **Single Node Assumption:** This study models Luzon and Visayas grids as individual single nodes. This means that line losses and transmission congestion are not accounted in this study.
2. **Weather Data Resolution:** The renewable generation profiles are derived from historical data from the MERRA-2 reanalysis dataset [19]. This does not model future climate change impacts. Furthermore, solar and wind profiles are simulated at the centroid of each CREZ [20].
3. **Weighted Spatial Aggregation:** The grid-wide renewable generation profiles are constructed using a weighted average based on the technical potential of each CREZ. This assumes that future capacity expansion will be distributed across zones strictly in proportion to their theoretical resource potential.
4. **Battery Storage Operational Assumptions:** For the Energy Duration Adequacy Test, the Battery Energy Storage System (BESS) fleet is modeled as a generic, energy-constrained resource with a fixed 4-hour duration [13]. The model assumes operational flexibility, allowing batteries to discharge at partial power to extend duration during long deficits.

Chapter 4

Methodology

This section will discuss the development and implementation of this study as summarized in Figure 4.1.

The methodology of this study will consist of three parts: (i) data collection and processing; (ii) model building; and (iii) simulation and analysis.

The Data Gathering and Processing stage will focus on sourcing data and how they will be processed into relevant inputs. Additionally, this subsection will discuss the calculation of the composite values relevant for the calculation of the hourly net load.

The Model Building stage, on the other hand, will discuss the multi-decadal net load profiling as well as the establishing of the recurrence matrix and visualizing it into a heatmap. The net load fingerprint will be tested against Philippine Power Statistic 2023 [21] to verify the model.

Finally, the planned dependable capacities will be simulated and tested against the net load profile provided in the model. This will also identify the gap, if any, between the net load and projected capacities in PDP 2023-2050.

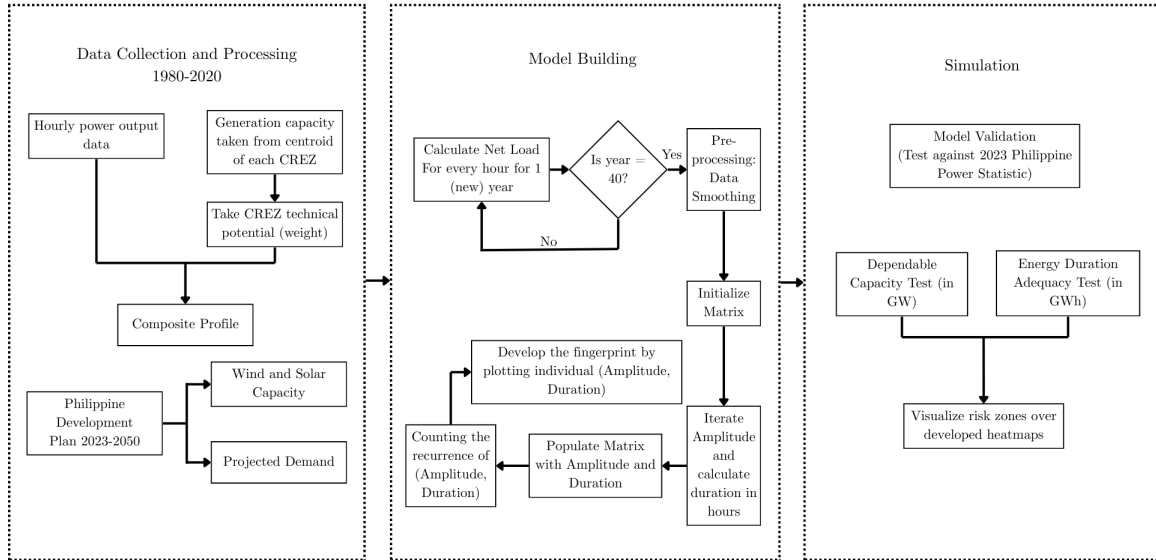


Figure 4.1: Methodology Overview Flowchart

4.1 Data Gathering and Processing

The data parameters and the corresponding sources are summarized in Table 4.1.

Table 4.1: Data Parameters and Sources

Data Parameters	Sources
Forecasted Demand	Power Development Plan 2023–2050 [4]
Build Capacities Per Technology	
Weather Data	Renewables.ninja [22]
Capacity Factor	
CREZ Capacities	Grid Planning and Competitive Renewable Energy Zones (CREZ) in the Philippines [20]
CREZ Coordinates	Renewable Energy Data Explorer [23]

This study will utilize the data presented in the Department of Energy’s Power Development Plan (2023–2050) [4], specifically focusing on the generation capacities outlined in the Reference Scenario, CES 1, and

CES 2 for the years 2025 through 2050 at five-year intervals. Peak Demand Forecast from the same plan will also be used.

For modeling renewable energy generation, historical hourly power output data (1985–2024) are sourced directly from Renewables.ninja [22][24][25]. Instead of manually calculating physical profiles from raw weather data, this study utilizes the built-in “Virtual Wind Farm” and “Virtual PV” algorithms of the Renewables.ninja platform as it uses validated MERRA-2 dataset and high-precision algorithms [19][24]. This will account for complex variable interactions such as temperature-dependent efficiency losses for solar and wind speed and hub height for wind. It is important to note that this study will take the centroid of the Competitive Renewable Energy Zone [20] as the coordinate input in Renewables.ninja.

The CREZ coordinates are given in Table 4.2

Table 4.2: Centroid Coordinates of Competitive Renewable Energy Zones (CREZ) [23]

CREZ	Island Name	Centroid Latitude	Centroid Longitude
Luzon			
L1	Luzon	15.3932	120.7703
L2	Luzon	15.0321	120.6782
L3	Luzon	15.3911	120.2106
L4	Luzon	15.6640	120.7696
L5	Luzon	16.2025	120.0192
L6	Luzon	18.2852	120.8563
L7	Luzon	18.1048	121.7055
L8	Luzon	17.5620	121.7050
L9	Luzon	16.7485	121.6106
L10	Luzon	13.7691	122.4435
L11	Luzon	14.0396	122.4445
L12	Luzon	13.8576	122.8148
L13	Luzon	12.9499	123.7340
Mr1	Mindoro	12.9580	121.1468
Mr2	Mindoro	12.5984	121.1474
Visayas			
N1	Negros	10.8809	122.9828
N2	Negros	10.6107	122.8897
Py1	Panay	11.4231	122.8023
Py2	Panay	10.9721	122.8000
S1	Samar	12.2135	125.1096
B1	Bohol	9.9682	124.2561

Two factors will be considered in developing the composite profile:

1. **Technical Potential (Capacity):** This is based on the weights of each CREZ. It is calculated as shown below:

$$w_{r,i} = \frac{z_{r,i}}{\sum_{r \in R} z_{r,i}} \quad (4.1)$$

Where:

$w_{r,i}$: is the weight of an individual Competitive Renewable Energy Zone of one source (i.e., solar or wind) in a single grid (i.e., Luzon or Visayas)

$z_{r,i}$: is the capacity of a single zone (e.g., $L1$, $L2$)

2. **Hourly Normalization:** This is based on the variability of the generation of each source (solar or wind) per hour. To facilitate the hourly normalization, a 1 MW Reference System is simulated for the centroid of each Competitive Renewable Energy Zone (CREZ). The simulation parameters are standardized as follows:

- **Solar PV:** System Loss = 10%, Tilt = 10°, Azimuth = 180° (South), Capacity = 1 MW [20]
- **Wind:** Hub Height = 100 m, Turbine Model = Vestas V90 2000, Capacity = 1 MW [20]

Finally, using the weights of each CREZ and the hourly data from Renewables.ninja, the hourly composite of each RE will be computed.

$$g_{\text{composite},p,h} = \sum_{r \in R} (g_{r,h,i} \times w_{r,i}) \quad (4.2)$$

Where:

$g_{\text{composite},p,h}$: is the hourly composite value for a specific source (solar or wind) in one grid (Luzon or Visayas)

$g_{r,h,i}$: is the hourly capacity factor based on weather conditions for a specific source (solar or wind) in one grid (Luzon or Visayas)

This unitless normalizing factor will then be used to scale the target capacities set by the PDP. The raw data above will be processed and collated using Python as relevant inputs to the calculation of the net load.

4.2 Model Building

This subsection will focus on generating the net load duration curves for each weather-year from 1985-2024. In addition to this, the development of the recurrence matrix and the subsequent heatmap visualization will also be discussed.

4.2.1 Net Load Profile

One of the objectives of this study is to develop a net load profile and to characterize the full net load variability. To achieve this, the study will adapt the net load formula by Ullmark et al [7].

$$n_{y,h} = d_{pdp} - [(g_{composite,w,t} \times c_{pdp,w}) + (g_{composite,s,t} \times c_{pdp,s})] \quad (4.3)$$

Where:

$n_{y,h}$: is the net load for the specific weather-year for 1 hour

d_{pdp} : is the projected demand for one year (i.e. 2030, 2035, ..., 2050) from the Power Development Plan 2023-2050

$g_{composite,w,h}$: is the hourly composite value for for wind

$c_{pdp,w,h}$: is is the installed capacity target for wind set in PDP

$g_{composite,s,h}$: is the hourly composite value for for solar

$c_{pdp,s,h}$: is is the installed capacity target for solar set in PDP

This will be iterated to span 8,760 hours, representing a full weather-year. This will be visualized as the load duration curve for one weather-year to highlight the system's peak capacity requirements. The same algorithm will be applied to all 40 weather-years.

4.2.2 Recurrence Matrix and Heatmap Development

In representing the net-load variability, the recurrences for 40 weather-years of the net load events will be enumerated for each amplitude and duration. This follows a two-step algorithm to develop this matrix.

1. **Data Smoothing:** A 12-hour rolling average will be applied to the hourly load profile. This is to filter short-term, low-energy fluctuations to focus on significant multi-day energy deficits. However, the data will also be analyzed when the 12-hour rolling average is not applied.

2. **Event Counting:** The matrix will be populated by identifying continuous net-load periods. To create a continuous fingerprint, this study will also apply the modified event counting by Gorranson [26] where the algorithm will count each net-load as both its maximum duration and as each discrete duration between zero and the maximum. Figure 4.2 represents the heatmap development.

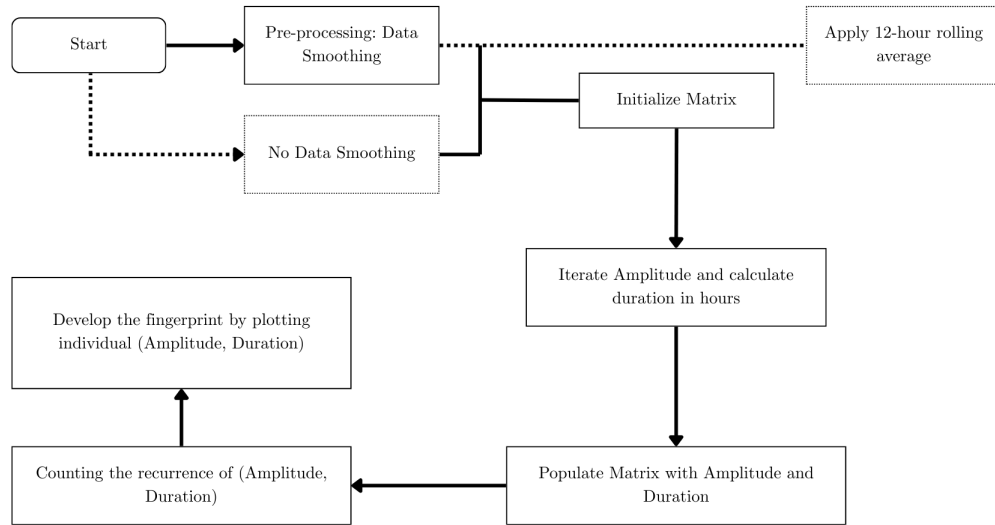


Figure 4.2: Flowchart of Heatmap Development

The resulting matrix will be visualized into a heatmap where the x-axis will represent the amplitude, the y-axis will represent the duration, and the z-axis will be the color scale to represent the frequency or the recurrence of the net load duration for the entire 40 weather-years. Provided in Figure 4.3 is the heatmap in Figure 3 from Ullmark et al [7].

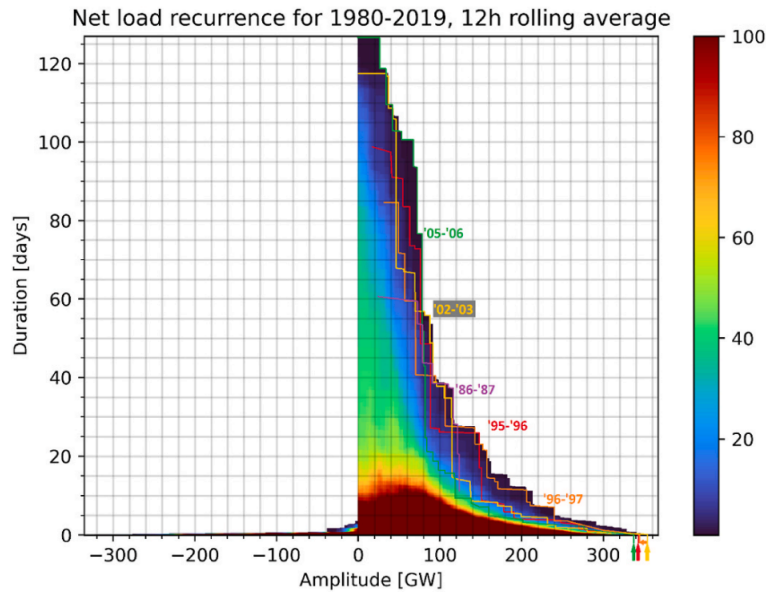


Figure 4.3: Heatmap of the Recurrence [7]

4.3 Simulation and Analysis

In assessing the sufficiency of the generation capacity set in the Power Development Plan 2023–2050, the developed net load duration and variability curves will be tested against dispatchable generation. The analysis will consist of two distinct tests.

4.3.1 Dependable Capacity Test

This test will determine the sufficiency of engines to cover the drops in renewable generation. To perform this analysis, the values in the x-axis (Amplitude in GW) will be subtracted from the dependable capacity target. If the result is less than zero, then there is a capacity gap during those extreme events.

4.3.2 Energy Duration Adequacy Test

For the purpose of the Energy Duration Adequacy Test, this study assumes that the Battery Energy Storage System (BESS) fleet operates as a flexible resource. This means that this approach assumes that the BESS can discharge the battery slowly for longer durations. This test will determine the sufficiency of batteries (in GWh) to outlast a continuous renewable deficiency. For the purpose of this analysis, this assumes battery

systems with a storage duration of four hours. This assumption is based on NREL Storage Futures Study, which establishes that 4-hour storage is sufficient to meet the integration of needs of high renewable energy penetration [13]. To perform this analysis, a duration curve will be developed and be tested against battery capacities. This curve is defined as:

$$\tau_{\text{limit}}(x) = \frac{c_{\text{pdp,batt}} \times t_{\text{rated}}}{x} \quad (4.4)$$

Where:

τ_{limit} : is the maximum sustainable duration (y – axis value). This will also define the duration curve

x : is the net load amplitude (x – axis value)

$c_{\text{pdp,batt}}$: is the Total Installed Battery Capacity in GW

t_{rated} : is the Rated Duration Assumption, in this case – 4 hours

This curve represents the maximum sustainable duration for any given power deficit amplitude. Historical net-load event located above this curve indicates insufficient storage capacity, signifying that while the instantaneous power capacity may be sufficient, the system lacks the stored energy volume to sustain generation throughout the continuous deficiency.

4.3.3 Visualization

To visualize the results, specific areas will be determined and will be drawn over the net load heatmap, where:

1. **Capacity Risk Zone:** The right of the vertical axis will be called Capacity Risk Zone where the PDP lacks in dependable capacities.
2. **Energy Risk Zone:** This is the duration curve mentioned in 4.3.2. The value above this curve is the Energy Risk Zone where the batteries cannot handle such long-term events.
3. **Safe Zone:** The area between the duration curve and the vertical axis are the events that the dependable capacity targets can handle easily.

An example of this is shown in Figure 4.4. Note that this is only to represent the zones and are not actual results from the study by Ullmark.

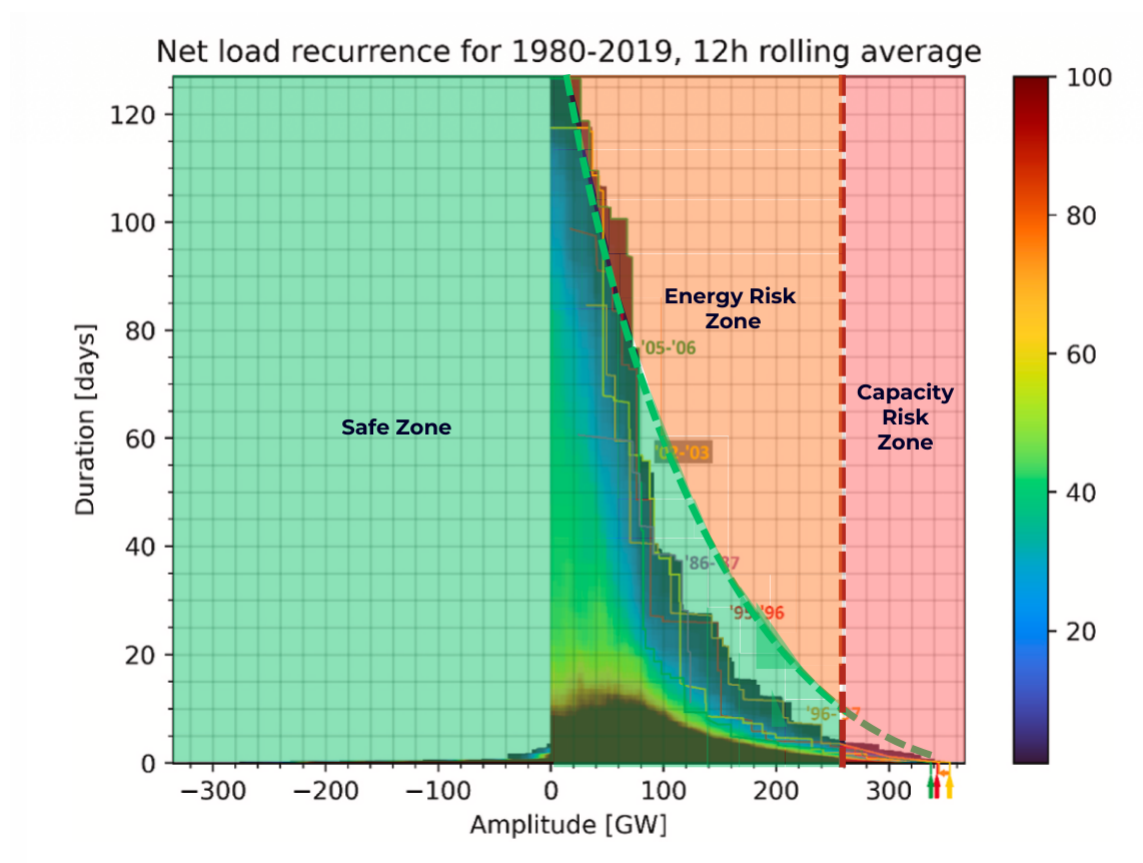


Figure 4.4: Figure 3 from Ullmark et al with “Capacity Risk Zone”, “Energy Risk Zone”, and “Safe Zone” Determined

Chapter 5

Preliminary Findings

5.1 Data Gathering

The researchers have completed substantial preliminary work including weather data acquisition, and PDP scenario compilation. This section details the progress achieved in preparation for the next research phase.

5.1.1 Weather data

The weather data for this project has been obtained from Renewables.ninja. Specifically, 16 years (1985-2000) of weather data have been extracted for 10 of the 21 CREZ in Luzon and Visayas grids designated by DOE.

5.1.2 PDP Reference Scenario

The DOE's PDP 2023-2050 specifies capacity expansion requirements under three scenarios, Reference Scenario, CES1, and CES 2. For this project, all of these scenarios have been adopted.

First, the Reference Scenario aims for a renewable energy share of 35% in the generation mix by 2030 and 50% by 2040. For the Reference Scenario, specific capacity expansion schedules for Luzon and Visayas grids have been compiled from the PDP for six planning horizons: 2025, 2030, 2035, 2040, 2045, and 2050. The PDP specifies committed and projected renewable energy capacity additions by technology.

Table 5.1 summarizes the Reference Scenario renewable energy capacity expansion for Luzon and Visayas

grids across the planning horizons. The table shows the progression of solar PV and wind capacities specified in the PDP for each planning year.

Table 5.1: Reference Scenario: Installed Capacity Targets in GW [4]

Technology	2025	2030	2035	2040	2045	2050
Luzon Grid						
Solar	5.17	8.72	14.58	21.23	24.89	30.74
Wind	1.34	3.24	5.6	8.35	10.45	13.2
Visayas Grid						
Solar	1.43	3.39	6.14	8.89	13.76	13.84
Wind	0.64	2.23	4.89	7.55	10.73	10.77

5.1.3 PDP CES 1

Clean Energy Scenario 1 (CES 1) aims for a diversified generation mix by integrating nuclear energy alongside the renewable energy targets. For CES 1, specific capacity expansion schedules for Luzon and Visayas grids have been compiled from the PDP for six planning horizons: 2025, 2030, 2035, 2040, 2045, and 2050. The PDP specifies committed and projected capacity additions, characterized by the inclusion of nuclear power technologies.

Table 5.2 summarizes the CES 1 capacity expansion for Luzon and Visayas grids across the planning horizons. The table shows the progression of renewable energy and nuclear capacities specified in the PDP

Table 5.2: CES 1 Scenario: Installed Capacity Targets in GW [4]

Technology	2025	2030	2035	2040	2045	2050
Luzon Grid						
Solar	5.33	6.99	12.48	21.11	26.95	33.04
Wind	1.33	2.61	5.36	8.11	13.11	16.11
Visayas Grid						
Solar	0.79	1.76	3.91	6.91	10.90	12.40
Wind	0.90	2.94	4.94	6.94	8.94	10.14

for each planning year.

5.1.4 PDP CES 2

Clean Energy Scenario 2 (CES 2) aims for a high-ambition decarbonization pathway by maximizing the deployment of offshore wind resources. For CES 2, specific capacity expansion schedules for Luzon and Visayas grids have been compiled from the PDP for six planning horizons: 2025, 2030, 2035, 2040, 2045, and 2050. The PDP specifies committed and projected capacity additions, prioritizing increases in offshore wind installations.

Table 5.3 summarizes the CES 2 capacity expansion for Luzon and Visayas grids across the planning horizons. The table shows the progression of offshore wind and other variable renewable capacities specified in the PDP for each planning year.

Table 5.3: CES 2 Scenario: Installed Capacity Targets in GW [4]

Technology	2025	2030	2035	2040	2045	2050
Luzon Grid						
Solar	5.33	7.75	11.02	13.37	22.19	28.69
Wind	1.83	4.16	6.36	7.59	10.34	12.54
Visayas Grid						
Solar	0.79	1.56	3.06	3.96	4.21	4.21
Wind	0.90	3.16	3.16	3.16	3.16	3.16

Chapter 6

Project Schedule and Deliverables

6.1 Gantt Chart

This project is scheduled to be completed throughout the Second Semester of the Academic Year 2025-2026, covering a 16-week period from January to May 2026. However, these dates may be adjusted to align with any changes to the UP Diliman's academic calendar.

The project schedule, detailed in Figure 6.1, is divided into four distinct phases:

1. **Data Gathering and Data Processing (Weeks 1–2):** The initial phase involves the continuation of data gathering and the processing of relevant inputs using Python. These foundational tasks involve all team members.
2. **Model Building (Weeks 3–12):** During this period the team splits responsibilities to run parallel grids modeling. Both researchers follow an identical timeline:
 - (a) **Reference Scenario:** Developing net load profiles (Weeks 3–5) followed by the development of their recurrence matrices and heatmaps (Weeks 6–7).
 - (b) **CES 1:** Developing net load profiles (Weeks 8–9) followed by the development of their recurrence matrices and heatmaps (Week 10).
 - (c) **CES 2:** Developing net load profiles (Weeks 10–11) followed by the development of their recurrence matrices and heatmaps (Week 12).
3. **Simulation and Analysis (Weeks 7–14):** Following the modeling phase, the team:

(a) **Reference Scenario:** Simulates dependable capacity and energy duration adequacy tests (Weeks 7–8).

(b) **CES 1:** Simulates dependable capacity and energy duration adequacy tests (Weeks 9-10).

(c) **CES 2:** Simulates dependable capacity and energy duration adequacy tests (Weeks 11-12).

Results gathering overlaps with the weeks of model building and testing (Weeks 8–12), followed by result analysis (Week 13–14) and the drawing of conclusions (Week 13–14).

4. **Finalization (Weeks 15–16):** The final weeks are dedicated to finishing the manuscript, delivering the final presentation, and submitting the final manuscript.

Finally, The writing of manuscript is maintained throughout the 16 weeks.

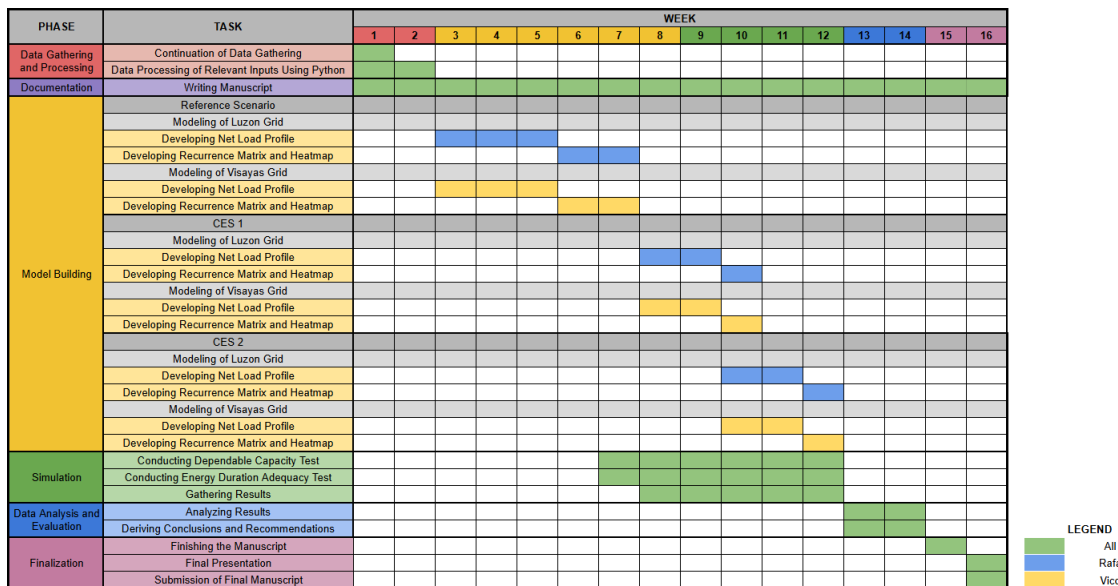


Figure 6.1: Gantt Chart

6.2 Project Deliverables

6.2.1 Halfway-point Deliverables

This project aims to accomplish the following mid-point deliverables by Week 8:

1. Gather the data needed for Luzon and Visayas grids.
2. Process the relevant inputs using Python.

3. Develop net load profile of the grids individually for Reference Scenario.
4. Develop recurrence matrix and heatmap of the grids individually for Reference Scenario.
5. Start developing the net load profile of the grids individually for CES 1.

6.2.2 Final Deliverables

By the end of the 16th week, the project is aiming to complete the following deliverables:

1. Repository containing all data inputs and output files generated for every established model.
2. Simulation results to analyze, identify, and quantify any adequacy gap between the PDP's planned capacity and the system's reliability requirements.
3. Slide deck and poster for final presentation.
4. Short paper and appendices that document the project specifics.

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